

Linux kernel gets fix for 11-year-old root hole in DCCP code

Linux developer Andrey Kononov has released a fix for an 11-year old bug in the Linux kernel. The security hole is in the support for Datagram Congestion Control Protocol (DCCP) that was introduced in 2005.

The flaw can be exploited by malicious software on a vulnerable device or gain root-level access when users log into their accounts. Once reached through a backdoor, attackers can leverage the vulnerability to compromise the system and even acquire a box from a connected network or Internet. Moreover, the programming blunder is in how DCCP code handles a socket buffer (skb).

According to the email list announcement by Kononov, an skb for a `DCCP_PKT_REQUEST` packet is forcibly freed via `__kfree_skb` in `dccp_rcv_state_process` if the `dccp_v6_conn_request` is returned successfully. An attacker can then gain access and control what object that would be, and even rewrite its content with arbitrary data. If the object has any triggerable functions, the attacker can execute the arbitrary code in the kernel. "An attacker can control what object that would be and overwrite its content with arbitrary data by using some of the kernel heap spraying techniques," explained Kononov.

The fix has been released for the Linux community to reduce instances of the DCCP flaw. It is recommended to update your system as soon as your distro gets the patch. Meanwhile, you can remove the buggy DCCP support from your kernel to avoid the impact of the vulnerability.

Microsoft Azure gets Kubernetes support

Microsoft has announced that it has added Kubernetes support to its Azure Container Service. The new development has expanded Azure with the inclusion of three major orchestration technologies, namely, Windows Server Containers, Kubernetes and DC/OS — rivaling other public cloud platforms.

Kubernetes is one of the leading orchestrators in the world of containers. Though Google initially promoted the open source container cluster manager, companies like Baidu, Canonical and Intel have recently added Kubernetes support. And now, it is Microsoft that has favoured Kubernetes for Azure.

Microsoft's Azure team launched the preview support for Kubernetes in November last year. That preview launch helped the Redmond company improve the support, and ultimately led to its general availability. "With this news, we again deliver on our goal of providing our customers the choice of open source orchestrators and tooling that simplifies the deployment of container-based applications in the cloud," said Saurya Das, program manager II, Azure Linux.

The Kubernetes support on Azure Container Service comes weeks after Docker debuted on Microsoft's cloud platform. The newly formed combination will upgrade container management on Azure and give developers a reason to prefer Microsoft over Amazon and Google.

Along with bringing the Kubernetes support, Microsoft has also opted for the preview of Windows Server Containers to provide enterprise customers an additional choice in orchestrator using the Azure Container Service. The Azure team has also updated the existing DC/OS support to version 1.8.8 to deliver an improved experience.

You can deploy an Azure Container Service cluster using the Azure portal or Azure CLI 2.0, and experience Kubernetes or other orchestrators including Windows Server Containers or DC/OS.

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Linux 4.10 brings support for virtualised GPUs

Linus Torvalds has finally brought out the anticipated Linux 4.10 release.

The new Linux kernel is specifically designed to enhance graphics on high-end hardware and thus, you can find native support for virtual GPUs.

Coming after eight release candidates, Linux kernel 4.10 is the result of nearly 13,000 commits that the Linux community received from its members all around the world. This is the reason Torvalds considers it as a 'fairly average release' and not as small as it initially seemed to be. "On the whole, 4.10 did not end up as small as it initially looked," the 47-year-old Torvalds wrote in a mailing list announcement.

One of the most noticeable changes that you can find in the final Linux 4.10 release is the arrival of support for virtual GPUs. This means that you can now have an immersive experience when using virtual GPUs on an updated Linux system. Additionally, the latest kernel includes improved support for GPU models and some special treatments for AMD Radeon.

Apart from improving support for GPUs, Linux 4.10 incorporates several improvements to file systems as well. You can see some of the instances on EXT4, F2FS, XFS and OverlayFS. The new Linux kernel also includes a new hybrid block polling method that reduces the load on CPU.

Linux 4.10 introduces some major changes for mobile devices. There is support for newer Qualcomm Snapdragon chips and compatibility with Allwinner A64. The kernel also includes certain driver upgrades and better caching options.

"I expected things to be pretty quiet, but it ended up very much a fairly average release by modern kernel standards," Torvalds stated.

Linux 4.10 would take some time to reach your favourite distro. However, you can download the new update from the official kernel.org website.